1. $0 \le x < 55 \quad 7x + 11 \equiv 35 \bmod 55$
   gcd(7, 55) = 1 ∴ an inverse exists
   the inverse, a, is found through the form:
   $7a \equiv 1 \bmod 55$
   ∴ $a = 8$

   $(7)(8)x + (11)(8) \equiv (35)(8) \bmod 55$
   $x + 11 \cdot 8 \equiv 35 \cdot 8 \bmod 55$
   $x \equiv 8(35-11) \bmod 55$
   $x \equiv 192 \bmod 55$
   ∴ $x \equiv 27 \bmod 55$

2. $0 \le x < 247 \quad x \equiv 5 \bmod 19 \quad x \equiv 11 \bmod 13$
   Using the Chinese remainder theorem, we have m = 19, n = 13, a = 5, b = 11
   To find sm + tn = 1 plug in the values for m and n which gives:
   $S19 + t13 = 1 \quad s = -2$ and $t = 3$

   Plugging these values into the Chinese remainder theorem equation:
   $x \equiv a + (b - a) \cdot sm \bmod mn = b + (a - b) \cdot tn \bmod mn$
   $x \equiv 5 + (11 - 5) \cdot -2 \cdot 19 \bmod 19 \cdot 13$
   $x \equiv -223 \bmod 247$
   ∴ $x \equiv 24 \bmod 247$

3. $0 \le x < 55 \quad 0 \le y < 55 \quad 2x - 2y \equiv 49 \bmod 55 \quad 3x + y \equiv 11 \bmod 55$

   First eliminate y by multiplying $3x + y \equiv 11 \bmod 55$ by 2 yielding:

   $6x + 2y \equiv 22 \bmod 55$ and add this equation to $2x - 2y \equiv 49 \bmod 55$ yielding:

   $8x \equiv 71 \bmod 55$

   $8x \equiv 16 \bmod 55$

   gcd(8, 55) = 1 ∴ an inverse exists
   the inverse, a, is found through the form:
   $8a \equiv 1 \bmod 55$
   ∴ $a = 7$

   Plugging the inverse back in:

   $x \equiv 16 \cdot \bmod 55$

   $x \equiv 112 \bmod 55$

   ∴ $x \equiv 2 \bmod 55$

   Substituting the x value of 2 back into the equation $3x + y \equiv 11 \bmod 55$ to get the value of y:

$3x + y \equiv 11 \bmod 55$

$3(2) + y \equiv 11 \bmod 55$

$\therefore y \equiv 5 \bmod 55$